

SECURITY AUTOMATION & ORCHESTRATION (SOAR) LAB

Building an Automated Incident Response Pipeline with Tines & VirusTotal

Author:	Dominique King	Environment:	Tines Cloud & VirusTotal API
Focus Area:	SOC Automation & Threat Intel	Date Completed:	July 15, 2026

1. Executive Summary & Lab Objectives

Modern Security Operations Centers (SOCs) are constantly inundated with security alerts, many of which involve repetitive analysis of Indicators of Compromise (IoCs) such as IP addresses, file hashes, and domains. To combat analyst fatigue and reduce Mean Time to Resolution (MTTR), security teams deploy Security Orchestration, Automation, and Response (SOAR) pipelines.

The objective of this hands-on lab is to design, implement, and validate an automated SOC orchestration workflow using **Tines** and **VirusTotal**. The workflow simulates a real-world pipeline where an external threat detection system triggers a webhook with an IP address, which is automatically enriched via the VirusTotal threat intelligence engine, and subsequently formatted into an actionable SOC alert emailed directly to security analysts.

Security Engineering Value Proposition:

This project demonstrates advanced cloud integrations, secure API management, structured JSON payload parsing, and dynamic variable templating. These skills are highly sought-after in modern security engineering and automated incident response roles.

2. High-Level Architectural Flow

The SOAR pipeline consists of three primary, interconnected stages:

- Ingestion (Webhook):** A custom endpoint designed to receive inbound HTTP POST payloads containing raw alert data (e.g., an IP address requiring investigation).
- Enrichment (HTTP Request):** A REST API connector that queries the VirusTotal v3 database dynamically using the parsed IP address and secure credentials.
- Notification (Send Email):** A dynamic template engine that maps the retrieved threat intelligence variables (status, classification, URLs) into a cleanly formatted notification email dispatched to analysts.

3. Detailed Step-by-Step Implementation

Step 1: Webhook Ingestion Setup

An ingress point was constructed utilizing the Tines Webhook action. This block generates a unique, cloud-hosted URL configured to listen for incoming JSON payloads. To model a live firewall or SIEM detection alert, a test payload was structured containing a targets parameter:

```
{  
  "ip_address": "8.8.8.8"  
}
```

Step 2: API Integration & Key Isolation (VirusTotal)

An HTTP Request action block was integrated downstream from the Webhook. For security best practices, the VirusTotal API Key was securely isolated. A temporary static test was configured using a known IP address ([8.8.8.8](#)) to confirm connectivity and authenticate the handshake:

```
https://www.virustotal.com/api/v3/ip_addresses/8.8.8.8
```

The HTTP method was set to **GET**, and the headers were securely mapped using the authorization parameters. Execution verified successful communication with a standard [HTTP 200 OK](#) payload.

Step 3: Transitioning to Dynamic Variable Mapping

To ensure the pipeline is completely modular and scalable, the static test URL was replaced with a dynamic variable pattern utilizing Tines' liquid templating syntax. The endpoint URL was reconfigured to programmatically fetch the specific IP address provided by the triggering webhook payload:

```
https://www.virustotal.com/api/v3/ip_addresses/{{webhook.body.ip_address}}
```

Step 4: Creating the Dynamic Alert notification

A **Send Email** block was connected downstream from the HTTP Request action. Using Tines' native data mapping, raw JSON keys returned from VirusTotal were mapped into user-friendly "pills." The template was designed to provide analysts with immediate actionable indicators:

```
Hi Team,  
  
The automated SOC pipeline has detected suspicious activity.  
  
Details of the investigated IP:  
- IP Address: {*webhook.body.ip_address*}  
- Scan Status: {*http_request.body.data.type*}  
- VirusTotal Link: {*http_request.body.data.links.self*}  
  
Please review the details in the VirusTotal console.
```

4. Portfolio Artifacts & Screenshot Mapping Reference

The following table outlines the structured screenshots captured during this lab deployment, serving as objective verification of technical competency for the portfolio:

#	Screenshot Filename	Description & Key Validation Elements
1	1_Webhook_Setup.png	Webhook action block initialization on the Tines canvas. Displays generated unique endpoint URL and structural configuration.
2	2_Webhook_Payload.png	Webhook execution logs showing the initial ingress payload parsed from a mock threat trigger (e.g., {"ip_address": "8.8.8.8"}).
3	3_VT_API_Action.png	HTTP Request configuration settings demonstrating GET method and secure header setup.
4	4_VT_Hardcoded_URL.png	Demonstrates temporary static sandbox testing with hardcoded destination to verify credentials.
5	5_VT_API_Error_Fix.png	Capture of the active session debugging process showing successful API communication and key error elimination.
6	6_VirusTotal_Data_Expanded.png	Expanded JSON schema returned by VirusTotal. Validated by highlighting HTTP Status 200 and ID 8.8.8.8 parsing.
7	7_Tines_SOAR_Workflow.png	Complete bird's-eye logical design of the SOAR Storyboard, detailing Webhook -> HTTP Request -> Send Email.
8	8_Tines_Email_Log_Success.png	In-platform debug log confirming successful processing and packaging of the email block with status "success".
9	9_SOC_Alert_Email.png	Final validation artifact: The actual alert email delivered directly to devonk21@gmail.com with dynamically populated variables.

5. Key Security Engineering Takeaways

- **API Integration Mastery:** Hands-on experience constructing, authenticating, and debugging RESTful APIs within security tooling.
- **Structured Output Parsing:** Demonstrated competency in reading nested JSON arrays and extracting valuable metadata dynamically.
- **Operational MTTR Reduction:** Moving from standard manual copy-paste analysis of IoCs to dynamic, sub-second SOC alerting workflows.